AVATAR · early surgery vs watchful waiting in asymptomatic severe aortic stenosis · outcome: all-cause death

RoB 2 asks up to 22 questions in 5 domains. Results describe risk of bias.

Y yes · PY / PN probably yes / no · N no · NI no information ▨ not needed (RoB 2 branching)

Evidence key: yellow = quote directly supports the answer · grey = background context · dashed = reasoning without a quote

Domain & question	The sources say	AI model answers	ARBITER (rule-based software, no AI) checks and applies the rules	Result
Randomisation process Was allocation concealed until participants were assigned? 1.2	**Paper p 3** "... web-based interactive response system ... computer-generated ... permuted-block sequence with variable block size."	**Yes** Central web-based allocation.	✓ Quote found in stored paper Open point recorded: whether the system was run independently of enrolling staff is not stated. 1.1 Y · 1.2 Y · 1.3 N	**Low** RoB 2 rules
Deviations from intended interventions Did participants know which treatment they received? 2.1	**Registry** "Masking: NONE"	**Yes** Open-label surgery vs watchful waiting.	✓ Found in the stored registry record 2.1 Y · 2.2 Y · 2.3 PN · 2.4 ▨ · 2.5 ▨ · 2.6 Y · 2.7 ▨ 3 questions not needed (RoB 2 rules)	**Low** RoB 2 rules
Missing outcome data Were outcome data available for nearly all participants? 3.1	**Paper p 6** "All patients were included in the analysis. ... One patient ... lost to follow-up ... censored..."	**Yes** 1 of 157 (0.6%) incomplete.	✓ Quote found in stored paper 3.1 Y · 3.2 ▨ · 3.3 ▨ · 3.4 ▨ RoB 2 rules → Low	**Low** RoB 2 rules
Measurement of the outcome Did outcome assessors know the treatment? Could that have influenced how deaths were recorded? 4.3, 4.4	**Paper p 3** "Data and safety monitoring board members were not blinded to the treatment allocation during events review."	**Yes** → **Probably no** *All-cause death involves no judgement.*	✓ Quote found in stored paper (4.3) 4.4 rests on reasoning, recorded as such (no quote). 4.1 PN · 4.2 PN · 4.3 Y · 4.4 PN · 4.5 ▨	**Low** RoB 2 rules
Selection of the reported result Was the analysis plan fixed before the results were known? 5.1	**Paper p 4 · context** "Prespecified secondary end points included the following: ... Time to death" **Registry · context** "Primary outcome 1: all cause death, Major Adverse Cardiac Event (MACE) including: ..."	**No information** The paper says "prespecified", but no dated plan lists this analysis; the registry lists death only inside a combined outcome.	✓ Both quotes found in the stored sources 5.1 NI · 5.2 PN · 5.3 PN RoB 2 rule: plan not shown to be fixed in advance; no sign of selective reporting → Some concerns	**Some concerns**

Overall ARBITER applies the RoB 2 rule: any domain with Some concerns → overall Some concerns — **Some concerns**

The AI model never sets the rating — ARBITER computes it from the answers with the official RoB 2 rules

Every quote is checked against a stored, unchangeable copy of the source

When the sources are silent, the answer says so — No information, not a guess

Without ARBITER

The same AI model, with the same instructions but without ARBITER, rated every domain Low. On question 5.1 it wrote:
"Time to death was a prespecified secondary endpoint with a registered 5-year time frame." — no source given; the registry lists death only inside a combined outcome.

Expert adjudicators (both): Some concerns

✓ ARBITER ✗ Without ARBITER

Published review: Low

Everything above is saved in one audit file (.arbiter.zip); an independent verifier checked it: verified.